

Cascadia: Resident 975B MoE Inference on Eleven AI PCs

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Abstract

Mixture-of-experts models make nearly trillion-parameter capacity accessible with sparse per-token computation, provided that the serving system can distribute the weights and coordinate their execution. We present Cascadia’s resident execution of Inkling, a 975B-total/41B-active-parameter model, on eleven Intel Core Ultra X7 358H AI PCs, each with 64 GB of memory, Arc B390 integrated graphics and gigabit Ethernet. The work contributes a compressed execution architecture that fits six consecutive decoder layers per machine, a streaming pipeline that coordinates prompt admission and concurrent decoding, and a method for evaluating draft heads against states and token IDs captured from the deployed numerical path. A unified fused-expert representation accommodates both sparse and dense feed-forward layers, reducing measured dense-layer call time from approximately 8.1 to 4.5 ms. Paired measurements at fifteen concurrency levels from 1 to 176 streams reach 60.29 aggregate decode tokens/s at 88 streams, with 46.87 tokens/s over the complete serving phases. At fifteen streams, median first-token latency is 6.05 s. Evaluation on captured fleet states separates the effects of vocabulary selection and weight quantization on draft agreement. Together, these contributions establish an execution and evaluation approach for large sparse models on distributed client systems with shared CPU–GPU memory.

1 Introduction

The sparse activation of mixture-of-experts (MoE) models changes the relationship between parameter capacity and computation. Inkling has 975B total parameters but activates 41B per token [21]. Serving such a model requires access to its expert weights even though each token selects only a subset. A collection of AI PCs offers a useful combination of distributed memory capacity, integrated matrix-compute accelerators and general-purpose CPU cores. Realizing that combination requires a serving architecture designed around shared CPU–GPU memory and sequential autoregressive dependencies.

Cascadia distributes Inkling’s text decoder across eleven Intel Panther Lake machines. Each machine holds six consecutive decoder layers and executes its own portion of the model on the local CPU and integrated GPU. Compressed projections and fused expert operations execute on the GPU, while the CPU manages attention state, routing and stream coordination. Independent requests occupy different pipeline stages concurrently. A direct token-return connection closes the autoregressive loop, and bounded prefill windows allow prompt processing to progress through the same chain.

The research contribution is the design and evaluation of this integrated serving system at nearly trillion-parameter scale on 64 GB client devices. The work develops three aspects that are closely coupled in this setting:

1. **Resident execution across shared-memory accelerators.** We combine layer partitioning, operator-specific compression and CPU–GPU placement to keep the model distributed across the fleet. A common fused-expert representation supports both sparse MoE blocks and the model’s dense feed-forward blocks. The measured dense-layer implementation takes approximately 4.5 ms per call, compared with 8.1 ms for three compressed matrix operations.
2. **A streaming pipeline for concurrent sparse-model service.** We integrate per-stream attention and convolution state, balanced admission, eight-row prefill windows and a direct sampled-token return path. A paired concurrency curve from 1 to 176 streams characterizes the resulting service. The highest measured paired mean occurs at 88 streams, eight per pipeline group: 60.29 aggregate decode tokens/s and 46.87 whole-phase tokens/s. Fifteen streams yield 6.05 s median first-token latency.
3. **Draft evaluation aligned with deployed inference.** We capture FP32 residuals together with the fleet’s emitted token IDs and replay each draft module’s temporal state. On 36 sequences, the resulting analysis separates vocabulary selection from additional weight quantization: restricting the head to 65,536 tokens changes first-draft agreement by 2.271 percentage points, while quantization at that vocabulary changes it by a further 0.175 percentage points.

These contributions extend Cascadia’s earlier pipeline-shard and architecture work [4, 19]. The new setting combines a much larger sparse model, compressed expert residency, dense/sparse operator unification and evaluation on distributed quantized states. We position the individual mechanisms relative to consumer-cluster inference, MoE execution and speculative decoding in Section 7.

2 System architecture

2.1 Panther Lake as a shared-memory inference platform

Intel’s Core Ultra X7 358H provides sixteen cores and threads: four performance cores, eight efficient cores and four low-power efficient cores. Its Arc B390 GPU has twelve Xe cores and uses the Xe3 graphics architecture. The processor supports LPDDR5X up to 9600 MT/s and includes an NPU rated at 50 INT8 TOPS [11, 12]. Cascadia uses the CPU and integrated GPU for this deployment. CPU state processing and GPU matrix operations draw on the same local memory capacity and bandwidth.

The fleet’s aggregate capacity is distributed across independent address spaces. Model partitioning assigns weights and sequence state to each machine; network transfers carry the intermediate

Component	Deployment configuration
Fleet	Eleven Core Ultra X7 358H machines, each with nominal 64 GB memory; 704 GB nominal capacity across the fleet.
CPU and GPU	Sixteen logical CPUs and an Arc B390 integrated GPU per machine.
Usable memory	Operating-system reports of 62,757 or 62,615 MiB per machine, approximately 61.15–61.29 GiB.
Runtime	Linux, with recorded kernel 7.0.0-31-generic; OpenVINO 2026.3.1 GPU execution and Rust serving/state management.
Network	Gigabit Ethernet; eight USB NCM interfaces and three PCIe-style interfaces.
Partition	Eleven pipeline roles, each owning six consecutive decoder layers.
Serving path	Text generation; deployment default of 1024 sequence positions, evaluated with short prompts and outputs.

Table 1: The hardware and execution configuration used in the study. Vendor accelerator and memory-rate specifications describe hardware capability; serving performance is measured separately.

activations needed by the next stage. The system therefore exploits local shared memory within a machine and explicit message passing between machines.

2.2 Model structure and partitioning

Inkling’s decoder has 66 layers. Its first two feed-forward blocks are dense, with intermediate width 24,576; the remaining 64 use routed and shared experts with intermediate width 3072. Each sparse layer selects six experts from a pool of 256 and also executes two shared experts. The hidden width is 6144. Attention combines sliding and global layers, learned relative-position bias and short convolutions [22].

Pipeline role 0 owns layers 0–5, token embedding and request admission. Roles 1–9 each own six intermediate layers. Role 10 owns layers 60–65, final normalization, vocabulary projection and sampling. Forward messages contain FP32 residuals, stream-slot identifiers and positions. A width-6144 residual occupies 24,576 bytes per row before message metadata. The final role returns sampled token IDs directly to role 0, which schedules the next decoding step.

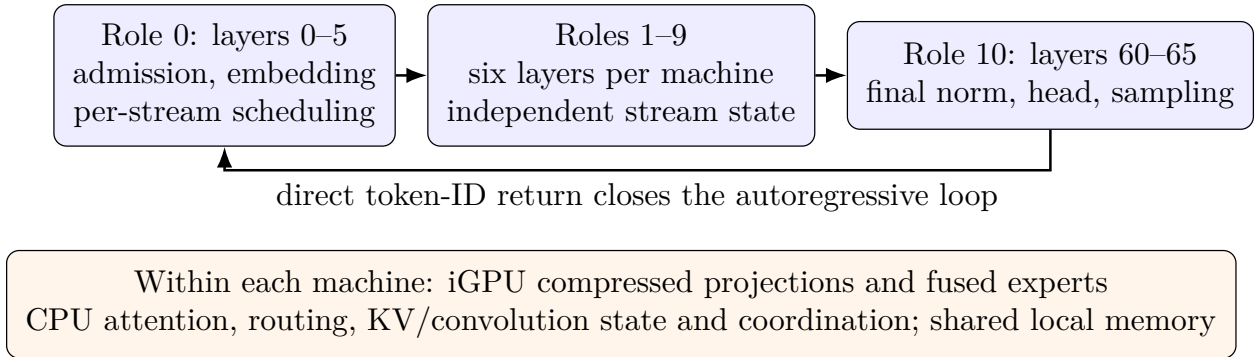


Figure 1: Cascadia’s resident streaming pipeline. Different requests can occupy different stages concurrently; consecutive ordinary tokens of one request depend on the sampled-token return.

3 Resident execution of dense and sparse layers

3.1 Compression and placement as one design

The execution architecture assigns numerical formats by operator. Expert weights use group-32 four-bit quantization; attention projections and the output head use eight-bit weights. OpenVINO’s fused-expert path performs half-precision arithmetic, while the Rust runtime maintains routing decisions, attention and convolution history, and the FP32 residual stream.

This placement keeps the GPU’s compressed expert representation resident while avoiding a second large resident CPU expert cache in normal operation. The configuration budgets approximately 52 GiB for GPU pages on each machine, with a larger allocation on the final role. Per-layer scaling of the expert up-projection and corresponding output compensation manage the range of the fused arithmetic. The relevant allocation unit is the entire layer shard: expert storage, projections, sequence state and host working memory share one physical capacity budget.

The contribution is a resident execution architecture for this model and client-memory setting. It combines local placement with distributed partitioning: the cluster supplies weight capacity, each integrated GPU supplies compressed projection/expert execution, and CPU cores maintain the stateful operations that connect successive tokens. Recorded profiles include fallback counters alongside GPU timings, so residency and execution placement remain observable properties of the implementation.

3.2 A common fused representation for dense and sparse blocks

The two dense feed-forward blocks can use the same fused-expert operator family as the sparse layers. Partitioning the intermediate dimension into eight slices of width 3072 gives

$$W_d[\text{SiLU}(W_g x) \odot W_u x] = \sum_{j=1}^8 W_{d,j}[\text{SiLU}(W_{g,j} x) \odot W_{u,j} x]. \quad (1)$$

Every slice is active with unit weight. The dense block thus matches the backend’s expert width without introducing a learned router or dropping intermediate activations. The algebra follows established dense-to-expert decompositions [18, 28]; our implementation maps that representation onto the compressed fused operator used by the resident Inkling shard.

Dense layer	Three matrix calls	Fused slices	Time reduction	Relative difference
Layer 0	8.15 ms	4.51 ms	44.7%	5.7×10^{-4}
Layer 1	8.11 ms	4.45 ms	45.1%	5.9×10^{-4}

Table 2: One-row load checks for the dense blocks. Relative difference compares the two FP16 execution paths. Equation 1 preserves the real-arithmetic function; the measured differences reflect finite-precision execution.

The two-row measurements are 8.41 ms per layer with the matrix-call path and 4.47/4.48 ms with fused slices. At the stage level, role 0’s observed computation changes from 51.9 to 43.7 ms per frame. These are operator and stage measurements; the complete serving rates reported in Section 5 characterize the distributed system separately.

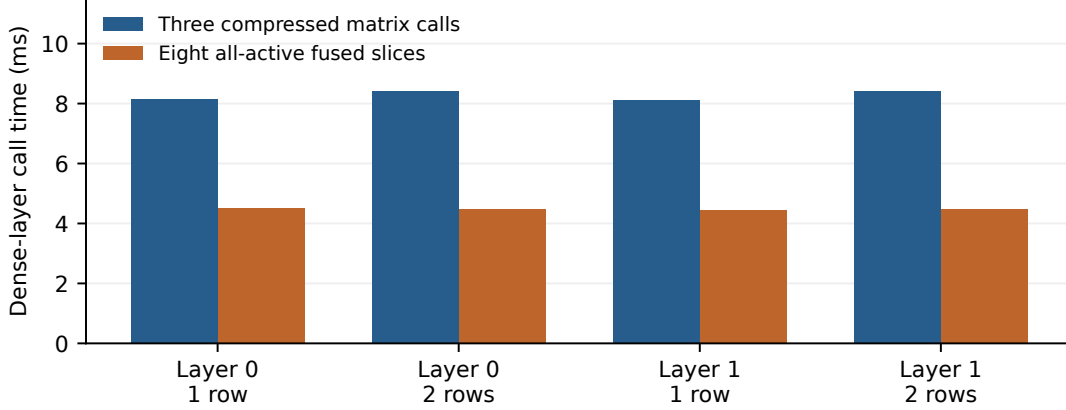


Figure 2: Dense-block execution using three compressed matrix operations or eight all-active fused slices. The operator improvement holds for the measured one- and two-row calls.

3.3 Weight traffic and the role of concurrent requests

Using the exported compressed sizes, an unshared full-model token accounts for approximately 16.76 GB of active expert weights, 8.59 GB of attention projections, 0.52 GB of dense weights and 1.24 GB of output-head weights: about 27.1 GB in total. These are logical weight bytes, including expert scale/zero-point overhead, rather than a count of physical DRAM transactions. Caching and reuse across rows determine how those bytes translate into memory traffic [24].

The layout makes two forms of parallelism explicit. Within a stage, a multi-row frame shares projection work and allows CPU attention to run across independent streams. Across stages, different frames use different memory buses concurrently. Consequently, aggregate service is a property of the distribution of requests over the pipeline as well as of the local kernels. This motivates the stream and admission design below.

4 Streaming service across the fleet

4.1 Stateful streams and direct token return

Each request occupies a stream slot with its own position, KV cache and convolution state. Frames group rows from independent slots, and each stage applies its local layers before forwarding the residuals. Role 0 admits work into the least populated group and advances the engine in short steps, allowing request submission to proceed while other frames are circulating.

The direct final-to-first connection carries sampled token IDs without requiring a reverse traversal through the intermediate workers. Receipt of an ID makes the next token of that stream schedulable. This explicit return path lets ingress interleave admission, forward progress and completion handling within the same serving loop. A full-chain readiness probe establishes downstream readiness before admission begins.

4.2 Prefill windows and parallel attention

Prompts enter as windows of at most eight rows. Consecutive windows can occupy different pipeline stages, and admission can proceed while ingress waits for token replies. The window size bounds the prefill work presented to a stage at one time. During decoding, independent rows provide parallel CPU work for attention even while the integrated GPU handles compressed projections and experts.

Chunked prefill and iteration-level scheduling have established precedents [1, 2, 27]. The contribution here is their realization within a resident sparse-model pipeline whose stages maintain both KV and convolution state, share memory between CPU and GPU, and communicate over gigabit Ethernet. Prefill, decode, row-parallel attention and direct token return are coordinated in one runtime.

4.3 Verified proposals for individual streams

For an individual active request, the serving loop can propose tokens using request history, a persisted phrase table and a small external draft model. The table is used for sufficiently confident contexts, with the model supplying other proposals. The target model verifies the proposed sequence; rejected positions rewind both KV and convolution state before decoding continues. This extends the existing speculative-decoding approach [4, 13] to Inkling’s stateful layer implementation.

The finalized survey evaluates twelve single-stream prompts twice, producing text-identical 128-token outputs for every pair. Aggregate decode rate over the serial requests’ decode intervals is 5.68 tokens/s in the first pass and 10.24 in the repeated pass; whole-phase rates are 5.19 and 8.77 tokens/s. Phrase learning remains active between passes, and capture instrumentation differs as described below. These paired observations characterize service with accumulated prompt history; they do not isolate a kernel speedup or estimate performance on unseen prompts.

5 Evaluation of concurrent serving

5.1 Measurement protocol

The finalized survey holds the model and serving binary fixed: group-32 INT4 experts, INT8 attention/head weights and the streaming configuration described above. Thirty mixed-workload phases cover fifteen concurrency levels from 1 to 176, with two phases per level and twelve deterministic prompt families. Each request uses greedy generation and emits 128 tokens. Requests within a cohort arrive approximately 15 ms apart; at low concurrency, successive complete cohorts supply at least twelve requests per phase. The complete survey also retains three explanation-family phases, for 33 measured phases, 1,592 requests and 203,776 generated tokens. Table 3 uses the thirty mixed-workload phases.

The paired passes reuse the same prompt templates. The main sweep proceeds upward and then downward in concurrency, with the 88-stream refinement measured as an additional pair. Phrase learning remains enabled throughout. Residual-capture writes are enabled at the ascending 1–15-stream points and disabled thereafter, including the repeated pass. The pairs therefore describe the serving system with its accumulated history and recorded instrumentation state. Within high-

concurrency cohorts, the finite prompt generator also produces some repeated inputs: the 88-stream phase contains 79 distinct prompts. The runtime release and binary hash, phase records and token traces accompany the artifact.

For cohort j , let f_i and l_i be the first and last token-event times of each request. The common decode interval is $(a_j, b_j]$, where $a_j = \max_i f_i$ and $b_j = \min_i l_i$. Let u_j count tokens emitted by all requests during that interval. We report

$$Q_{\text{decode}} = \frac{\sum_j u_j}{\sum_j (b_j - a_j)}, \quad Q_{\text{phase}} = \frac{\sum_i n_i}{t_{\text{end}} - t_{\text{start}}}. \quad (2)$$

Q_{decode} measures aggregate throughput while every request in a cohort is decoding. Q_{phase} includes admission, prefill, queueing and drain. Dividing Q_{decode} by concurrency gives the mean contribution per stream over these common intervals. Table and figure rates are arithmetic means of the two phase rates; TTFT quantiles pool the requests from both phases. The observed range of two phases is descriptive, rather than a confidence interval.

The client counts token-event multiplicities and checks them against each request’s final API token usage. The retained traces independently reconstruct all common intervals and whole-phase rates. TTFT refers to the first emitted model-token event, including reasoning and structural tokens. The collector checks server counters around each phase to isolate measured traffic. Software profiles supply frame-weighted stage timings, with physical device identity kept separate from pipeline role. Short serial/concurrent prefix checks and twelve known-answer questions provide functional validation; they are not a general model-quality benchmark.

5.2 Concurrency, throughput and first-token latency

Streams	Q_{decode}	Q_{phase}	Q_{decode}/C	TTFT median	TTFT p95
1	7.96	6.98	7.960	2.18	2.41
2	4.53	4.37	2.267	2.06	3.03
4	9.16	8.64	2.290	3.31	4.03
6	13.89	12.80	2.315	4.20	5.49
8	18.60	16.86	2.325	4.55	6.23
11	23.28	20.86	2.116	5.47	7.64
15	24.60	22.12	1.640	6.05	10.11
22	33.87	29.26	1.539	9.71	14.51
32	38.31	32.48	1.197	13.44	21.90
48	45.33	37.68	0.944	18.85	31.55
64	53.60	42.41	0.837	25.10	46.41
88	60.29	46.87	0.685	34.61	64.75
96	46.85	39.79	0.488	38.94	71.31
128	50.10	41.61	0.391	55.11	108.73
176	57.72	45.24	0.328	76.83	165.34

Table 3: Finalized mixed-workload concurrency curve. Every request produces 128 tokens. Rates are tokens/s; TTFT is in seconds. C is concurrent streams, rather than total requests at the low-concurrency points. Rates average two phases; TTFT quantiles pool their requests.

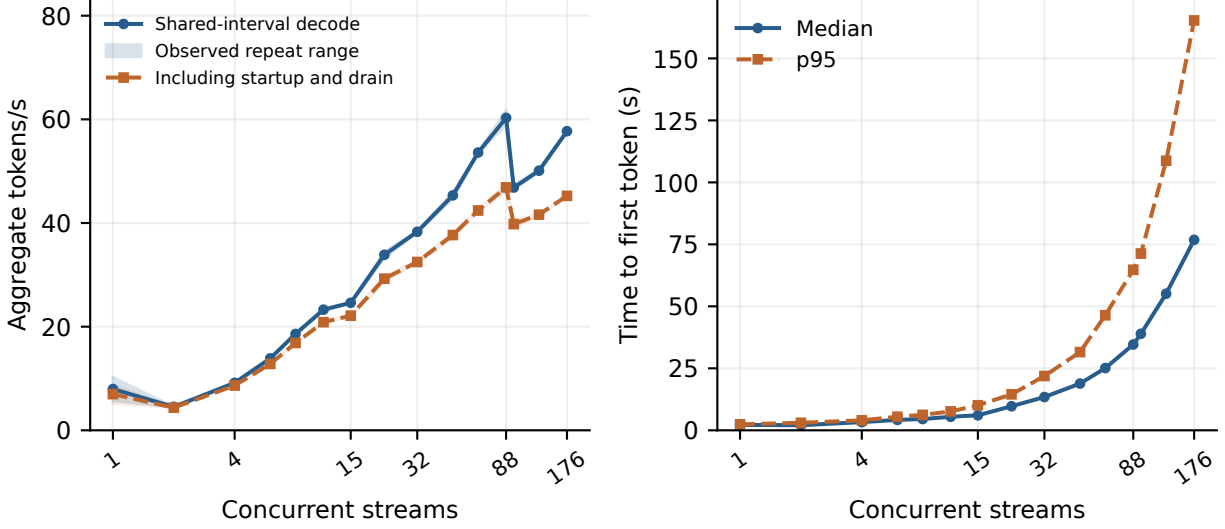


Figure 3: Paired measurements across fifteen concurrency levels. Left: common-interval decode throughput and whole-phase throughput, with shading showing the two observed phase rates. Right: pooled request TTFT median and 95th percentile. The 88-stream point is the highest paired mean on the measured grid.

At 88 streams, the two phases each generate 11,264 tokens. Common-interval decode rates are 58.86 and 61.71 tokens/s; whole-phase rates are 46.03 and 47.72 tokens/s. Their paired means, 60.29 and 46.87 tokens/s, are the highest on the evaluated concurrency grid for both measures. This operating point assigns eight streams to each of eleven pipeline groups. At 176 streams the fleet sustains 57.72 aggregate decode tokens/s and 45.24 whole-phase tokens/s, while fifteen streams provide 24.60 and 22.12 tokens/s with 6.05 s median TTFT.

The curve exposes a deployment-specific relationship between concurrency and the distributed execution schedule. The 88-stream alignment coincides with the highest observed throughput, and further concurrency redistributes work into larger per-group row counts. Throughput is nonmonotonic across that transition, while first-token latency grows as more requests enter the chain. These observations identify a useful measured operating point for this resident pipeline; they do not by themselves isolate the effects of row padding, expert routing and scheduling. The contribution is an empirical service envelope that connects the implemented architecture to explicit throughput and latency choices.

5.3 Effect of windowed admission

The reference fifteen-request burst configuration records median TTFTs of 31.52 and 31.23 s and whole-phase rates of 15.678 and 17.005 tokens/s. The windowed configuration records 6.91 s median TTFT and 22.521 whole-phase tokens/s. The median latency reduction is approximately 4.5-fold across these observed configurations.

This supporting configuration comparison uses distinct prompt tags and precedes the finalized fixed-binary survey. It characterizes the admission design; Table 3 characterizes the resulting deployment across concurrency levels. With one-second arrival spacing, the windowed configuration also records 2.48 s median TTFT for fifteen requests capped at 96 output tokens.

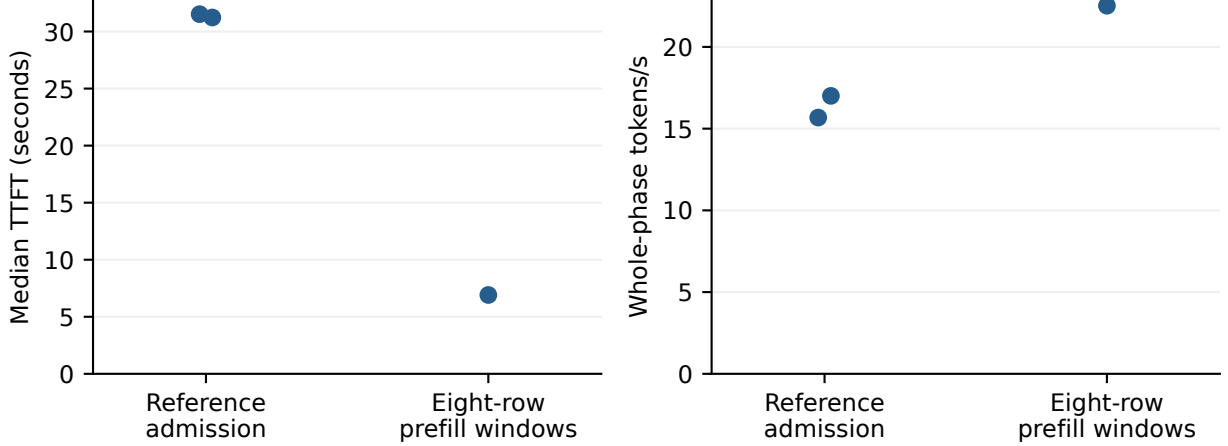


Figure 4: Fifteen-request bursts with an output cap of 128: reference admission and eight-row prefill windows. Each point is a retained phase. The comparison reports observed configurations with different prompt tags.

6 Draft evaluation on deployed states

6.1 Capturing the serving model’s actual computation

A draft head consumes internal representations whose values depend on the serving model’s quantization and execution path. We evaluate Inkling’s shipped multi-token prediction (MTP) head using states generated by the distributed fleet itself. The capture format records FP32 residuals together with positions, input token IDs and actual emitted token IDs. FP32 accommodates the observed pre-final-normalization range, which includes a sampled magnitude of 4,272,968.

The corpus consists of 36 sequences, three from each of twelve task families, with 160 generated positions per sequence. The 5,760 emitted tokens provide 5,724 first-draft prediction targets. All captured residuals pass finite-value checks, prompt token IDs match the rendered inputs, and decoded full responses match the completion API output. Actual emitted IDs define the evaluation labels, preserving the numerical decisions made by the deployed model.

Each of the eight MTP modules maintains its own attention and convolution history. Replay can supply true-token context or self-fed context while preserving the module-specific temporal state. This gives a controlled basis for studying the draft’s representation, vocabulary and numerical format independently of request-serving timing.

6.2 Separating vocabulary and weight-format effects

On the same captured fleet states, the original full head yields 66.81% first-draft agreement. Restricting the original head to a prefix of 65,536 vocabulary entries yields 64.54%. Applying INT4 feed-forward and INT8 projection/head weight grids at that vocabulary yields 64.36%.

Vocabulary selection accounts for 2.271 percentage points of the agreement difference, and additional weight quantization at the fixed vocabulary accounts for 0.175 percentage points. This decomposition identifies vocabulary size as the larger contributor in this specific deployment-format

Draft calculation on captured fleet states	First-draft agreement
Original weights, full head	66.81%
Original weights, 65,536-token head	64.54%
INT4/INT8 weight grids, 65,536-token head	64.36%

Table 4: Matched-state draft evaluation over 5,724 prediction targets. The weight-grid calculation uses FP32 arithmetic offline; the table measures agreement with emitted fleet IDs, not live speculative throughput.

comparison. It provides a concrete basis for treating vocabulary footprint and draft-weight precision as separate design choices.

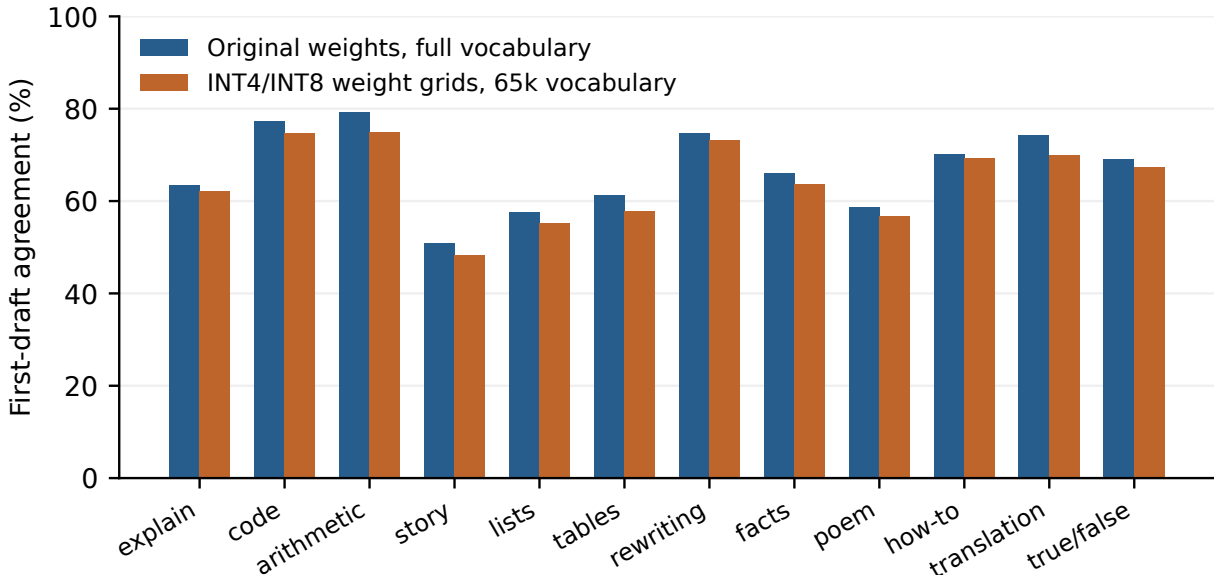


Figure 5: First-draft agreement by task family on captured fleet states. Each family contains three sequences. The two calculations share the same prediction targets and differ in draft weights and vocabulary.

The deployment-grid calculation records agreement of 0.746 for code, 0.748 for arithmetic and 0.482 for story. Across the eight-module chain, the reported expected number of accepted drafts is 2.128 with true-token context and 2.007 with self-fed context. These are descriptive offline measurements on the evaluated corpus. They complement the live phrase/model proposer results by exposing the draft head’s behavior on the distributed model’s actual states.

6.3 Contribution to draft evaluation

MTP and feature-based drafting are established approaches [9, 14]. MoE-specific speculative systems further connect acceptance, expert activation and verification cost [3, 10, 17]. Our contribution is an evaluation method and empirical decomposition tied to the numerical path of a distributed, compressed Inkling deployment: capture its residuals and emitted IDs, replay the draft’s full temporal state, and compare vocabulary and weight formats on those same labels. This connects model-level draft analysis to the exact serving configuration being studied.

7 Relationship to prior work

Client-cluster inference. Petals, TPI-LLM, prima.cpp and exo demonstrate inference across consumer or edge devices [5, 7, 15, 16]. Cascadia’s earlier work develops Intel/OpenVINO pipeline shards and the broader distributed architecture [4, 19]. The present work extends that lineage to a resident 975B sparse decoder on eleven 64GB integrated-GPU machines, including compressed expert placement and dense/sparse operator unification. A reported Inkling deployment on eight DGX Sparks uses tensor parallelism, NVFP4, MTP and a different memory/interconnect configuration [6]. The systems establish distinct deployment designs; the present evaluation uses the Intel fleet’s own workloads and numerical formats.

Sparse execution and operator representation. EdgeMoE and PowerInfer exploit sparsity and memory placement on resource-constrained hardware [20, 26]; Klotzki, MegaScale-Infer, EC2MoE and CPU/GPU hybrid serving examine expert-aware execution and distribution [8, 23, 25, 29]. Cascadia combines a resident layer partition with a CPU/iGPU split inside each shard. MoEfication and MLPMoE establish dense-to-expert decompositions [18, 28]. The measured contribution here is using all-active slices to map Inkling’s dense blocks onto the same compressed fused operator family as its sparse blocks on Arc B390.

Streaming and draft evaluation. Orca and SARATHI/Sarathi-Serve establish iteration-level scheduling and chunked prefill [1, 2, 27]. We integrate those principles with a gigabit sparse-model pipeline, short-convolution state and a direct sampled-token return path. The draft evaluation contributes captured-state measurements of the deployed model, complementing speculative decoding, MTP and MoE-specific verification research [9, 10, 13]. The evidence supports these architectural, implementation and empirical contributions without requiring a priority claim for the underlying algebra or scheduling primitives.

8 Conclusion

Cascadia demonstrates resident inference of Inkling’s 975B MoE decoder on eleven Panther Lake AI PCs. The system combines compressed expert placement, a unified fused representation for dense and sparse blocks, and stream-oriented coordination across integrated GPUs sharing memory with CPUs. Its paired concurrency curve reaches 60.29 aggregate decode tokens/s and 46.87 whole-phase tokens/s at 88 streams, with 6.05s median first-token latency at fifteen streams. Captured-state draft evaluation further separates vocabulary and weight-format effects using labels emitted by the deployed model. These results provide an implemented architecture and measured techniques for serving large sparse models on distributed client hardware.

Artifact. The paper source, figures, derived tables and supporting evidence are available to reviewers in the private repository <https://github.com/labscommunity/cascadia-inkling-panther-lake-paper>. The evidence is pinned to Cascadia commit 3189a189fe3428f5a6315a7eb67b13148ec314e2. reproduction/CLAIMS.md maps the paper’s contributions and numerical statements to source records; the analysis scripts regenerate the reported tables and figures.

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